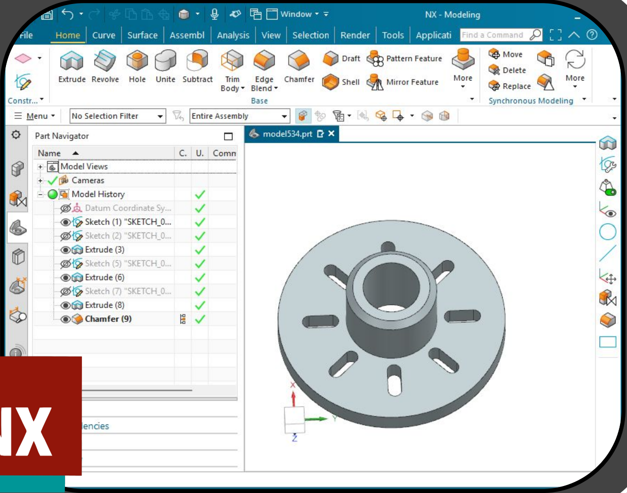
Skills Gained

Learned how to select an idea and properly write a NASA proposal by learning from NASA scientists, engineers, and managers. Additionally, chaired a proposal review session as a primary reviewer of another team's proposal by grading on criteria.

		Chair	
		Primary	
	Total Pages	Total for Proposal	
	7	5	
		Possible Score	
Technical Merit (50%)	3.5	5	
Proposal Objectives are clear	0.4		10
Viable Approach to development	0.4		10
Work plan well defined	0.4		10
Understands State of the Art	0.4		10
Identified the risks to the effort	0.4		10
Tasks described	0.4		10
Metrics for SOA Improvement Clear (e.g. KPPs)	0.9		20
Potential (Pass / Fail)			
NASA application identified		yes	
high payoff or clear infusion path		yes	
Improve cost, schedule, performance or risk?		yes	
Does it qualify for an NTR?		Yes	
Management Appraoch (30%)	2.1	5	
Team roles and responsibilities clear	0.4		10
milestones and deliverables clear	0.4		10
Good use of resources	0.4		10
Schedule credible / basis	0.4		10
Basis for costs clear	0.4		10
basis for skill mix / hours clear	0.4		10
Team / Work Force Development (10%)	0.7	5	
Skills / Capabilities of the key personnel	0.2		10
Facilities available for the scope	0.2		10
Availability of SME / Mentoring	0.2		10
Alignment (10%)		5	
Clear identification of NASA or National Need	0.7		10

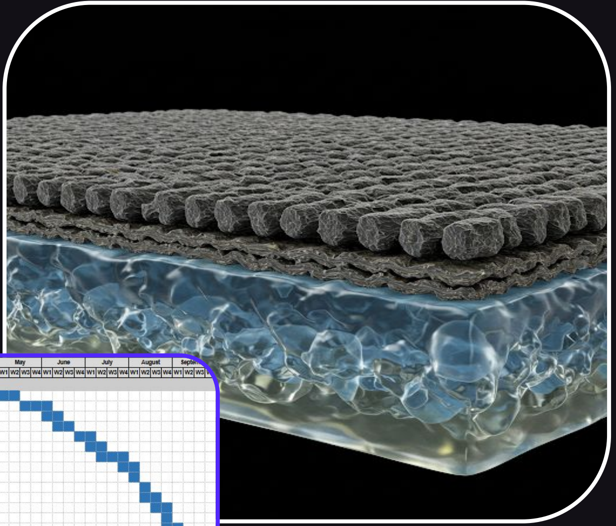
My team gained valuable experience by writing our proposal using the same evaluation constraints as other NASA proposals.

Got certified to use Siemens NX software to CAD 3D models. Learned the basic functions and operations to independently use the software.



Proposal Written

Our proposal addressed developing a Nanocellulose Hydrogel-Enhanced Whipple Shield (NCH-WS) to protect spacecraft from orbital debris with a focus on impact resistance and durability. The NCH in this innovation is able to reduce weight and increase vehicle lifespan.



	TASK TITLE					
1	Materials Research (May - October 2025)					
1.1	Review technical literature on spacecraft shielding materials.	Materials	May 5, 2025	May 16, 2025	10	13
1.2	Define performance targets for strength, healing, and environmental tolerance.	PI / Systems	May 19, 2025	Jun 2, 2025	10	13
1.3	Identify candidate nanocellulose hydrogel materials for testing.	CS / Materials	Jun 3, 2025	Jun 12, 2025	8	10
1.4	Coordinate with labs or vendors to obtain hydrogel samples.	PM	Jun 13, 2025	Jun 20, 2025	5	7
1.5	Synthesize nanocellulose hydrogel in lab settings.	CS	Jun 23, 2025	Jul 2, 2025	8	10
1.6	Evaluate chemical purity and composition of hydrogel samples.	CS / Materials	Jul 3, 2025	Jul 10, 2025	5	7
1.7	Create testable multi-layer shield samples with hydrogel, Kevlar, and aluminum.	Materials	Jul 11, 2025	Jul 22, 2025	8	10
1.8	Test tensile strength of hydrogel and composite materials.	Materials	Jul 23, 2025	Jul 29, 2025	5	7
1.9	Assess compressive resistance under load conditions.	Materials	Jul 30, 2025	Aug 1, 2025	3	4
1.10	Evaluate shear behavior of bonded materials.	Materials	Aug 4, 2025	Aug 6, 2025	3	4
1.11	Create micro-damages and measure hydrogel self-healing.	CS	Aug 7, 2025	Aug 13, 2025	5	7
1.12	Track recovery speed of hydrogel under standard conditions.	CS	Aug 14, 2025	Aug 20, 2025	5	7
1.13	Cycle samples through thermal extremes to simulate orbit conditions.	Materials / AE	Aug 21, 2025	Aug 27, 2025	5	7
1.14	Test hydrogel stability in vacuum conditions.	Materials	Aug 28, 2025	Sep 4, 2025	5	7
1.15	Expose hydrogel to radiation and monitor degradation.	CS / AE	Sep 5, 2025	Sep 11, 2025	5	7
1.16	Compile mechanical and environmental test data.	Materials	Sep 12, 2025	Sep 15, 2025	2	3
1.17	Analyze results to determine if samples meet design criteria.	Materials	Sep 16, 2025	Sep 18, 2025	3	4
1.18	Document validated material properties for design use.	PM	Sep 19, 2025	Sep 25, 2025	5	7
	Discuss test results in internal review meeting.	PI / PM	Sep 26, 2025	Sep 29, 2025	2	3

May W1 W2 W3 W4 | June W1 W2 W3 W4 | July W1 W2 W3 W4 | August W1 W2 W3 W4 | Septe… W1 W2 W3

K	L	M	N	O	P	Q	R	S		
Rating	Approach	Trend	Mitigation Plans	Status of Mitigation Plans	Trigger Date	Trigger Event	Closure Date	Closure Event		
High level indicator of the criticality of the risk	How the risk will be handled or not	Trend of any changes to the risk's LxC over its life cycle	Specific actions that will be implemented to mitigate the risk by either lowering the L and/or the C. Add or remove as many mitigation actions as needed and assign a new line and number (1,2,3...) to them. Carry this number across the next few rows where mitigation plans are monitored and controlled	Updates on the mitigation plans throughout their life cycle	The date that a mitigation plan should have an update by	A specific event that will be associated with the trigger date	The expected date the risk will be closed out	A specific event that will be associated with the closure date	The date followed by a short description of any changes to this risk Example: (10/22/2024): The risk's L was changed from a 2 to a 1 after completing mitigation plan #1.	copy/paste a summary of the risk into the risk matrix Rank, Risk ID #, Approach Note: This column does not need to remain when this risk log is being exported into any project documents
High	Mitigate (M)	↑ Increase	Conduct detailed impact testing on Kevlar and assess its performance at different velocities and temperatures.	Initial testing phase completed, waiting on final results.	45 Days after first application	Completion of impact testing under different velocity scenarios.	90 Days after trigger date	Successful impact testing showing Kevlar meets expected standards.		1-R1-M-↑
High	Mitigate (M)	→ Neutral	1. Conduct radiation degradation testing and explore protective coatings for Hydrogel to enhance its durability. 2. Explore advanced material coatings and conduct long-term degradation simulations.	1. Radiation testing initiated. 2. Coating material research initiated.	45 Days after first application	1. Completion of initial radiation exposure tests. 2. Completion of initial long-term degradation simulation.	120 Days after trigger date	Successful testing confirming Hydrogel's radiation resistance.		2-R2-M-→
High	Mitigate (M)	→ Neutral	Conduct thermal testing on material compatibility and adjust layer design for better expansion match.	Preliminary thermal testing complete.	30 Days after first application	Completion of thermal cycle tests on layered design.	90 Days after first application	Thermal cycling tests show good compatibility, adjusting design.		3-R3-M-→
High	Mitigate (M)	→ Neutral	Improve composite material testing and adjust thickness or configuration based on testing results.	Composite material tests ongoing.	30 Days after first application	Completion of first round of composite material testing.	90 Days after trigger date	Successful testing confirms shield's ability to absorb debris impact.		4-R4-M-→
Moderate	Mitigate (M)	↓ Decrease	Research alternative bonding techniques and perform adhesion strength tests.	Bonding strength tests underway, researching alternatives.	30 Days after first application	Completion of first round of bonding strength tests.	90 Days after trigger date	Confirmed adhesion standards met, material selection finalized.		5-R5-M-↓
Moderate	Watch (W)	↑ Increase								6-R6-W-↑
Moderate	Mitigate (M)	↑ Increase	Simulate integration tests in varied operational environments to confirm compatibility and make necessary adjustments.	Integration testing ongoing, monitoring for performance.	60 Days after first application	Completion of initial integration testing.	150 Days after trigger date	Integration test results confirm shield meets all mission requirements.		7-R7-M-↑
Moderate	Watch (W)	→ Neutral								8-R8-W-→
Moderate	Watch (W)	→ Neutral								9-R9-W-→

As the project manager for my team, my role was to address risk management, future timeline, budget, and connections with a Subject Matter Expert. Working as a team makes the intimidating task of proposal writing easier to manage and break down into small tasks.